



**WEAPONS
ROBOTICS AND
CONTROL
ENGINEERING**
UNITED STATES NAVAL ACADEMY

HOPPER INFORMATION SERIES

01 / HARDWARE AND SENSING

Hopper sensor suite

How an indoor quadrotor estimates motion and stays stable - without GPS.

NO GPS

INDOOR ONLY

SENSOR FUSION



The sensor map identifies every onboard sensing path



THIS HOPPER

- Uses IMU, barometer, magnetometer, optical flow, time-of-flight range, and camera sensing.
- Flies indoors using relative measurements and onboard sensor fusion.
- Does not include the optional GPS/GNSS receiver shown in the platform-family map.

NO GPS / GNSS

No latitude/longitude or global waypoint position.



What Hopper measures onboard

A IMU

3-axis accelerometer + 3-axis gyroscope: motion, tilt and angular rate.

M Magnetometer

A 3-axis digital compass used by the onboard system for heading cues.

P Pressure altimeter

Air-pressure changes provide a relative altitude trend to the controller.

F Flow sensor

Tracks apparent floor motion to help stabilize movement over textured surfaces.

T Time-of-flight

A short-range downward distance cue to the floor; performance depends on the surface.

C Camera

2 MP articulated camera: 45 deg forward-facing or 90 deg downward-facing.

I Color + IR array

24-bit color and infrared sensing provide optical/environment cues.

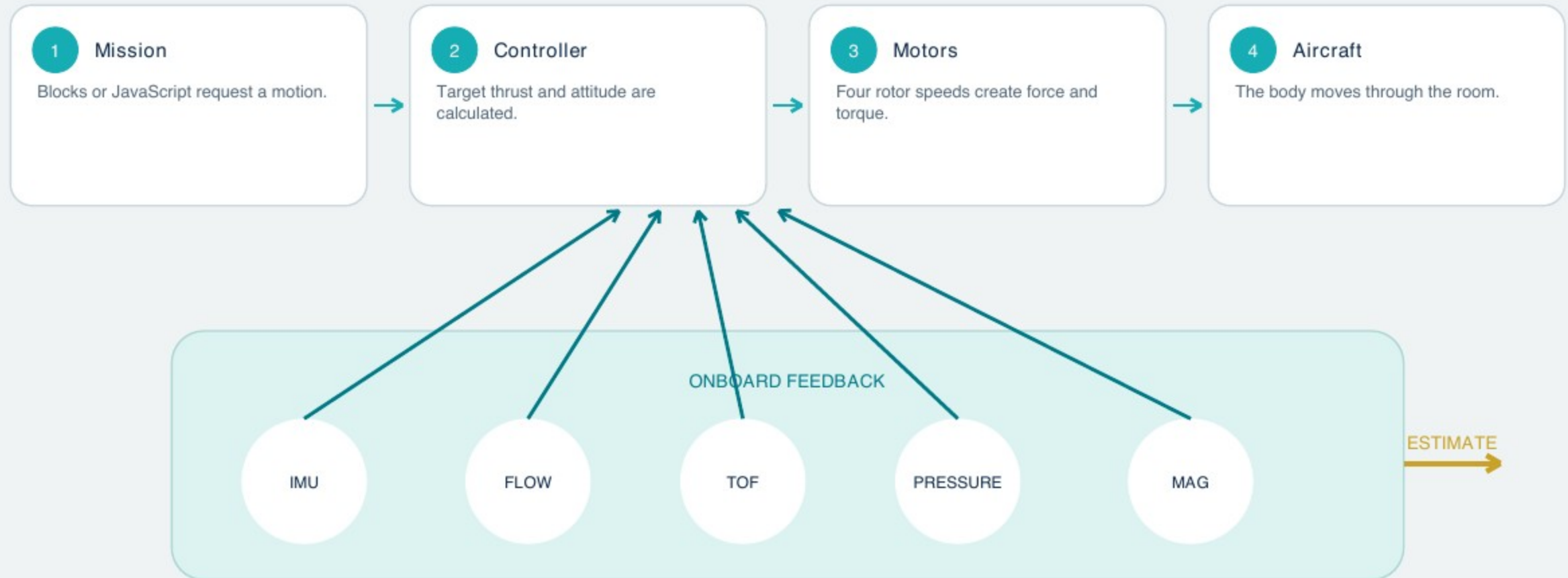
D Temperature

Ambient or onboard diagnostic temperature sensing - not a thermal camera.

Onboard sensor does not mean student-readable variable. Hopper Studio exposes only a subset.



Sensor fusion closes the control loop



Fast feedback is handled onboard; mission code sends higher-level commands.



No GPS is a design constraint

NO GPS

No latitude / longitude

No global waypoint position

Indoor use only

What it can do

- Stabilize attitude with onboard feedback.
- Estimate local motion relative to the floor.
- Use camera targets, objects, paper or AprilTags.

What student code cannot read

- No x/y room position or GPS coordinates.
- No real altitude, velocity or attitude value.
- No raw IMU, range or heading telemetry.

DESIGN MISSIONS AROUND RELATIVE CUES

- Time + power commands: fly forward for 1 second at 15%.
- Visual events: detect a white sheet, object label or tag ID.
- Conservative repeat-and-check logic instead of assumed position.

Camera and communication are separate paths



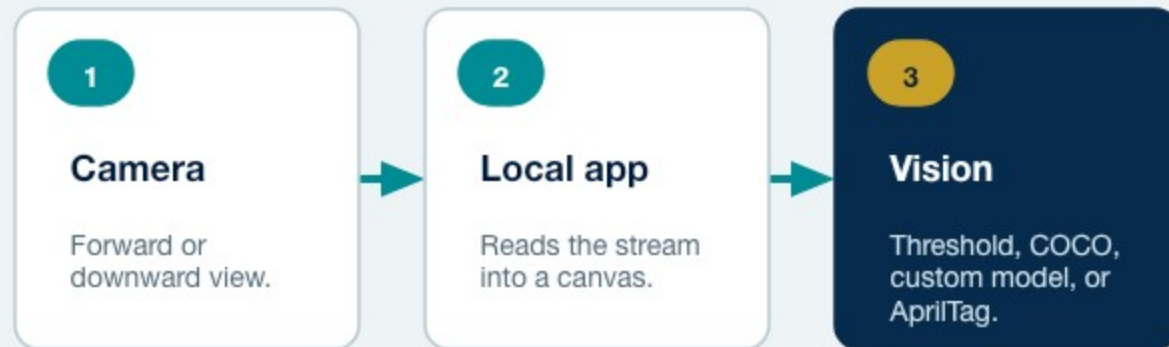
GPS / GNSS IS NOT INCLUDED

BLUETOOTH

- Flight commands.
- Battery and coarse flight state.

WI-FI / HTTP

- Camera video at 192.168.2.1.
- Vision needs readable image pixels.



A green Bluetooth link does not guarantee a readable camera feed.



Preflight for reliable sensing

ROOM + SURFACE

- Fly indoors in a well-lit room.
- Prefer gym or hardwood flooring with visible texture.
- Avoid strong vents, reflective concrete and difficult blue carpet.
- Take off from the floor, not from a hand or table.

APP + MISSION

- Check Bluetooth and Wi-Fi indicators independently.
- Test the exact target under classroom lighting.
- Use repeated observations before a flight decision.
- Keep three feet above obstacles and preserve a landing route.

SOURCES AND FURTHER READING

FTW Robotics - Hopper hardware and ideal flight conditions

<https://ftw-robotics.ai/hopper>

Hopper Studio - local implementation and classroom operating notes

README.md in the Hopper Studio project

Core rule: no GPS, indoor operation, relative cues, conservative decisions.